SAGE: SYNTHETIC AGING FOR A GRID ENVIRONMENT

SUPPLEMENTARY INFORMATION

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S1. COMPLETE PARAMETER SPECIFICATION

name	dtype	value	comment
alpha_cal	float	1.5	Scaling coefficient in calendar aging exponential term
beta_cal	float	0.75	Exponent for calendar degradation
beta_weather_cdd	float	0.5	Price increase per cooling degree-day
beta_weather_hdd	float	0.4	Price increase per heating degree-day
calibrated_temp_rise_c4	float	2	Steady state temp rise at C/4 discharge C
capacity_kwh	float	5000	Nameplate energy capacity kWh
cdd_shift_threshold	float	4	CDD threshold to classify as hot day
cold_day_peak_shift	int	-1	Hours to shift peak earlier on cold days
container_attenuation	float	0.0833	Fraction of outdoor swing that penetrates container
container_baseline	float	22	HVAC setpoint Â°C
default_gradient_factor	float	1	Default rack position for single runs
dispatch_window_end	int	21	Latest time of day for dispatch
dispatch_window_start	int	11	Earliest time of day for dispatch
E_a_cal	float	53000	Activation energy for calendar aging J/mol
E_a_cyc	float	35000	Activation energy for cycle aging J/mol
eta_discharge_bol	float	0.95	Discharge efficiency at BOL
eta_discharge_eol	float	0.9	Discharge efficiency at EOL
export_years	int	25	Number of years to export for TwinBESS
fleet_seed	int	43	RNG seed for fleet heterogeneity
fleet_size	int	1000	Number of modules in the fleet
gamma_spike	float	0.0025	Spike probability per hour
hdd_shift_threshold	float	4	HDD threshold to classify as cold day
hot_day_peak_shift	int	3	Hours to shift peak later on hot days
initial_soc	float	0.5	Initial state of charge when battery is turned on
k_cal	float	1.00E-05	Calendar aging base rate per hour
k_cyc	float	5.00E-05	Cycle aging base rate
measurement_noise_seed	int	123	RNG seed for sensor noise in exports
morning_peak_hour	int	7	Center hour of morning peak on cold days
morning_peak_magnitude	float	0.3	Height of morning peak as fraction of base price
morning_peak_width	float	2	Width of morning peak gaussian hours

name	dtype	value	comment
num_discharge_hours	float	4	Number of hours in a single discharge
P_dis	float	1000	Nameplate discharging power kW
pareto_shape	float	1.4	Spike shape parameter
phys_kelvin_offset	float	273.15	0C in Kelvin
phys_R_gas	float	8.314	Universal gas constant J/mol*K
phys_T_ref_kelvin	float	298.15	Reference temp for Arrhenius 25C
price_ceiling	float	5000	Max allowed price ERCOT offer cap
random_seed	int	43	Master RNG seed for reproducibility
seasonal_peak_day	float	200	Day of year for peak seasonal temperature peak
sigma_forecast_price	float	15	Standard deviation of price forecast error
sigma_manufacturing_quality	float	0.02	Standard deviation of battery quality
sigma_outdoor_forecast	float	1.5	Day-ahead outdoor forecast error Â°C
sigma_outdoor_noise	float	2	Weather noise std dev Â°C
sigma_soc_meas	float	0.02	SOC error std dev
sigma_soh_meas	float	0.01	SoH estimation error
sigma_temp_forecast	float	0.5	Day-ahead temperature forecast error C
sigma_temp_meas	float	0.5	Temp sensor noise C
sigma_temp_noise	float	0.5	Weather noise std dev C
simulation_start_date	str	1/1/2025 0:00	Simulation start timestamp
simulation_years	int	25	Duration of the simulation in years
SoC_max_BOL	float	0.95	Maximum allowed SOC at BOL
SoC_max_EOL	float	0.8	Maximum allowed SOC at EOL
SoC_min_BOL	float	0.05	Minimum allowed SOC at BOL
SoC_min_EOL	float	0.2	Minimum allowed SOC at EOL
SoC_ref	float	0.5	Optimal resting point where calendar degradation is minimized
SoH_EOL	float	0.7	SOH below which battery cannot be safely cycled
spike_base_magnitude	float	100	Base multiplier for Pareto spike magnitudes $/MWh
spike_cdd_multiplier	float	1.5	Spike probability multiplier on hot days
spike_hdd_multiplier	float	1.2	Spike probability multiplier on cold days
spike_temp_threshold	float	4	Degree-day threshold to trigger spike multiplier

name	dtype	value	comment
T_ambient_mean	float	22	Yearly average temperature C
T_amp_diurnal	float	0.5	Daily swing C
T_amp_seasonal	float	1	Seasonal swing +/- degrees C
T_BPT	float	22	Balance point where HDDs and CDDs are zero C
T_cell_max	float	55	Maximum safe operating temperature C
t_formation_hours	float	100	Hours from factory formation to site delivery
T_outdoor_amp_diurnal	float	6	Daily swing Â±Â°C (afternoon vs night)
T_outdoor_amp_seasonal	float	12	Seasonal swing Â±Â°C (summer vs winter)
T_outdoor_mean	float	22	Annual mean outdoor temperature Â°C
T_phase_shift_hours	float	14	Peak temperature hour (2PM)
unscheduled_outage_hourly_prob	float	0.0001	Hourly probability of unscheduled outage
vertical_temp_gradient_C	float	5	Temperature gradient bottom to top of rack C
viz_spike_threshold_multiplier	float	10	Multiplier for extreme price detection (VIZ-7)
viz_lifespan_cap	float	40	Maximum projected lifespan for sensitivity analysis (VIZ-8)
viz_cohort_high_percentile	float	75	Percentile threshold for high quality cohort (VIZ-4)
viz_cohort_low_percentile	float	25	Percentile threshold for low quality cohort (VIZ-4)
viz_position_threshold_high	float	0.75	Vertical position threshold for top rack cohort (VIZ-4)
viz_position_threshold_low	float	0.25	Vertical position threshold for bottom rack cohort (VIZ-4)
viz_loglog_exclude_years	float	0.5	Years to exclude from log-log regression early transient (VIZ-1)
viz_sqrt_exclude_years	float	0.1	Years to exclude from sqrt(t) fits (VIZ-1)
viz_hill_k_min	int	20	Hill estimator minimum k (VIZ-7)
viz_hill_k_max	int	500	Hill estimator maximum k (VIZ-7)
viz_hill_k_step	int	10	Hill estimator k step size (VIZ-7)
viz_efficiency_clip_min	float	0.8	Minimum efficiency for clipping (VIZ-3)
viz_efficiency_clip_max	float	0.99	Maximum efficiency for clipping (VIZ-3)
viz_revenue_display_start_year	int	12	Start year for revenue divergence plot (VIZ-3b)

Table S2a: Monthly Price Calibration Targets

Month	Mean Price ($/MWh)	Volatility (σ/μ)
January	30	1.2
February	28	1.1
March	25	1
April	30	1
May	35	1.2
June	40	1.3
July	55	1.7
August	65	2
September	45	1.5
October	35	1.4
November	30	1.1
December	32	1

Table S2b: Hourly Base Price Profile

Hour	Base Price ($/MWh)
0	45
1	40
2	38
3	35
4	35
5	40
6	55
7	75
8	70
9	65
10	60
11	55
12	50
13	50
14	55
15	70
16	90
17	110
18	120
19	115
20	95
21	80
22	70
23	60

Table S1 provides the complete parameter specification for SAGE simulations. All parameters are user-configurable through external configuration files; the values shown correspond to the baseline configuration used for the validation studies presented in the main text.

S2. PRICE MODEL CALIBRATION

The objective of the price model is not to reproduce wholesale market statistics or forecast market outcomes, but to generate physically plausible, temperature-correlated operational stress histories for degradation analysis. Monthly mean prices are explicitly calibrated, while higher-order statistics such as month-by-month volatility are treated as approximate, as they are strongly distorted by heavy-tailed scarcity events and the market offer cap. Prices are therefore used to induce realistic clustering and timing of dispatch rather than to model bidding behavior or revenue outcomes. Accordingly, the equations below define a controlled stochastic stress environment rather than a predictive or equilibrium market model.

S2.1 Ambient Temperature Model.

Ambient temperature (T_{amb}) is modeled as a superposition of deterministic seasonal and diurnal harmonics with stochastic weather variability. The hourly temperature $T_{amb}(t)$ is calculated as:

$$T_{amb}(t) = \bar{T} + A_{season} \cos\left(\frac{2\pi \cdot d(t)}{365}\right) + A_{diurnal} \cos\left(\frac{2\pi(h(t)-\phi)}{24}\right) + \epsilon_T(t) \qquad \text{(S1)}$$

where:

$\bar{T}$ is the annual mean temperature (Baseline: 22°C). A_{season} is the seasonal amplitude (Baseline: 10°C, typical of temperate climates). $A_{diurnal}$ is the diurnal amplitude (Baseline: 5°C). *d(t)* and *h(t)* represent the day of the year and hour of the day, respectively. ϕ is the diurnal phase offset, set to 14 (2:00 PM) to align peak temperature with typical daily maximums. $\epsilon_T(t) \sim \mathcal{N}(0, \sigma^2_T)$ represents stochastic weather noise (Baseline $\sigma_T = 0.5^\circ$ C).

While the noise term $\epsilon_T(t)$ is modeled as uncorrelated Gaussian noise, the dominant variation is driven by the deterministic components.

S2.2 Electricity Price Model

The synthetic electricity price generator is designed to capture a limited set of first-order statistical features of wholesale electricity markets relevant for operational stress generation, calibrated to approximate ERCOT-like conditions over the 2019–2023 period without relying on historical price replay. Prices are constructed using a small set of interpretable components that separately represent baseline structure, weather-driven demand effects, residual volatility, and rare scarcity events. The intent is not comprehensive market realism, but transparent control over the mechanisms that govern dispatch timing and exposure to high-stress operating events. All model parameters referenced in this section (e.g., γ_{spike}, α, β_{CDD}, β_{HDD}, p_m, and the diurnal multipliers D_h) are specified explicitly in a centralized parameter file and are read directly by the simulation at runtime.

From the perspective of degradation analysis, the individual price components play distinct and intentional roles. The deterministic backbone and weather-coupled terms primarily govern the frequency, seasonality, and clustering of dispatch, ensuring that daily operation is correlated with ambient temperature and long-term demand patterns. The residual volatility term introduces stochastic variation in dispatch timing within these constraints, while the scarcity spike process governs the occurrence of rare, high-price events that disproportionately influence realized revenue outcomes but do not determine dispatch timing, which is governed by forecast prices. The relative magnitudes of these components are therefore chosen to produce realistic operational stress histories rather than to enforce detailed agreement with market statistics.

*S2.2.1 Deterministic Backbone***.** A baseline price structure $p_{\text{backbone}}(m,h)$ captures standard seasonal and diurnal load patterns by combining monthly mean prices p_m with normalized hourly diurnal multipliers D_h:

$$p_{\text{backbone}}(m,h) = p_m \cdot D_h \tag{S2}$$

This term represents the expected price profile under typical operating conditions absent unusual weather or scarcity.

S2.2.2 Weather Coupling. To enforce realistic correlations between ambient temperature and electricity prices (e.g., elevated demand during heat waves or cold snaps), smooth weather-driven price deviations are modeled using cooling-degree-day (CDD) and heating-degree-day (HDD) sensitivities relative to a balance-point temperature T_{bal}:

$$p_{\text{weather}}(t) = \beta_{\text{CDD}} \max\ (T_{\text{amb}}(t) - T_{\text{bal}}, 0) + \beta_{\text{HDD}} \max\ (T_{\text{bal}} - T_{\text{amb}}(t), 0) \qquad \text{(S3)}$$

This term captures predictable, continuous price uplift associated with temperature-driven load changes.

S2.2.3 Residual Variability. Residual price variability and scarcity events are modeled explicitly to capture the heavy-tailed nature of operational stress rather than to enforce month-by-month volatility matching. Stochastic volatility introduces heteroscedastic fluctuations around the deterministic price structure, while scarcity events are represented using a Pareto-distributed spike process subject to an explicit offer cap. Because these rare events dominate the upper tail of the price distribution, realized monthly volatility is strongly influenced by spike timing and capping effects and is therefore not imposed as a strict calibration target. Instead, volatility emerges endogenously from the interaction of baseline structure, weather coupling, stochastic noise, and scarcity dynamics, consistent with the objective of generating realistic operational stress exposure rather than reproducing detailed market statistics.

Rare scarcity-driven price excursions are represented by an additive spike term $\Delta p_{\text{spike}}(t)$. Scarcity events are generated by a Bernoulli process with baseline probability γ_{spike}, specified in the parameter file: At each time step, the model makes a random yes/no decision about whether a scarcity event occurs. During hours classified as extreme hot or cold based on degree-day thresholds, the baseline spike probability is increased by fixed multipliers specified in the parameter file; during all other hours, the baseline probability is used unchanged. Conditional on a scarcity event, the spike magnitude is drawn from a Pareto distribution with shape parameter α and unit scale, with probability density function

$$f_X(x) = \alpha x^{-(\alpha+1)} \qquad x \geq 1 \tag{S4}$$

and is subsequently scaled into price units.

The Pareto distribution is a heavy-tailed power-law distribution commonly used to represent extreme events in economics and finance; smaller values of α produce heavier tails and more extreme price realizations. This formulation generates infrequent but large positive price excursions, consistent with the empirical upper-tail behavior of energy-only electricity markets.

S2.2.4 Market Price Construction. The market price time series is obtained by combining these components additively and enforcing the system-wide offer cap p_{ceiling} (baseline, $5,000/MWh):

$$p_{\text{market}}(t) = \text{clip}\,(p_{\text{backbone}}(t) + p_{\text{weather}}(t) + \varepsilon(t) + \Delta p_{\text{spike}}(t), 0, p_{\text{ceiling}}) \tag{S5}$$

This additive structure reflects the empirical behavior of energy-only electricity markets, in which scarcity conditions manifest as large, asymmetric price excursions superimposed on otherwise smooth baseline and weather-driven price dynamics, while preserving a clear separation between predictable structure, stochastic volatility, and extreme tail events.

S2.3 Price Forecast Model

To represent imperfect foresight without introducing a full market forecasting model, SAGE applies additive stochastic noise to the baseline price signal used for dispatch decisions. Specifically, forecast prices are generated by adding zero-mean Gaussian noise to the realized (or reference) price trajectory,

$$\hat{P}(t) = P(t) + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma_P^2) \tag{S6}$$

This abstraction introduces day-to-day uncertainty in the relative attractiveness of dispatch opportunities while avoiding explicit modeling of bid formation, market clearing,

or relative forecast error scaling. Forecast prices are used solely to rank dispatch opportunities and to select the highest-valued discharge block for each day, rather than to predict realized revenues or quantify forecast accuracy. Forecast errors are modeled as additive deviations because the objective is to induce imperfect dispatch ranking, not to replicate the statistical structure of real market forecasts.

S2.4 Block Scheduling Logic.

SAGE models discharge physics only; charging is treated as an exogenous boundary condition whereby assets begin each calendar day at their maximum achievable state of charge. This abstraction isolates the physical consequences of energy delivery from deployment-specific recharge logistics. Round-trip efficiency is estimated as $\eta_{RTE} = \eta_{discharge}^2$ (assuming symmetric charge and discharge losses).

Dispatch decisions are evaluated once per calendar day using a fixed-duration block strategy:

The scheduler identifies exactly one contiguous discharge block of prescribed duration (configurable via the num_discharge_hours parameter; baseline is 4 hours reflecting typical Resource Adequacy requirements).

- Block timing is selected to maximize the rolling average of forecast prices over the configured duration.
- Power is held constant throughout the discharge period.
- Blocks cannot span calendar day boundaries.
- No hour-by-hour re-optimization occurs within or after block commitment.

Degradation effects enter indirectly through power derating and energy availability, rather than through explicit cost terms. Power derating, if required, is determined by the physics engine at block initiation.

S2.5 Firm Power Determination.

When a discharge block begins, SAGE computes the maximum constant power sustainable for the full block duration:

$$P_{\text{firm}} = \frac{E_{\text{avail}} \cdot \eta_{\text{dis}}}{H} \tag{S7}$$

where E_{avail} is available energy within the current SOC window, η_{dis} is discharge efficiency, and H is block duration. Delivered power is the minimum of nameplate discharge power, firm-energy-limited power, and thermally-allowable power.

S3. DEGRADATION MODEL DERIVATIONS

This section provides the mathematical and numerical details underlying the degradation formulations introduced in the main text. The derivations here document how continuous time aging laws are discretized, stabilized, and evaluated under stochastic operating histories in the SAGE framework. No additional physical mechanisms are introduced; rather, the focus is on transparent numerical implementation, parameter interpretation, and consistency with the governing equations presented in the main text.

S3.1 Calendar Aging Implementation

While the main text defines the instantaneous degradation rate (Section 3.8.1), SAGE implements this numerically using a stress-weighted effective time formulation. This approach normalizes variable operating history into an equivalent duration at reference conditions:

$$q_{cal}(t) = k_{cal} \cdot (t_{eff}(t))^{\beta} \tag{S8}$$

where effective time t_{eff} accumulates according to:

$$t_{eff}(t) = \int_0^t f_T(T(\tau)) \cdot f_{SOC}(SOC(\tau))dt \tag{S9}$$

SAGE implements calendar aging using a stress-weighted effective time formulation that maps variable operating histories onto an equivalent duration at reference conditions. Effective time t_{eff} is a bookkeeping construct rather than an additional physical state variable; it accumulates monotonically as a function of thermal and state-of-charge stress while preserving the Arrhenius temperature dependence defined in the governing rate law. In this formulation, the power-law exponent β controls the temporal curvature of calendar aging with respect to accumulated effective time, while temperature sensitivity enters exclusively through the Arrhenius term. This separation allows variable operating conditions to be treated consistently without introducing chemistry-specific voltage or microstructural models.

While classical diffusion-limited models predict β = 0.5, meta-analyses of laboratory data report a wide range of fitted exponents across chemistries, test protocols, and state-of-charge conditions. For example, Attia *et al*[1]. demonstrated fitted values spanning approximately 0.16–0.55 across diverse datasets. This variability, together with the near-linear aging behavior frequently observed in grid-scale operation, motivates treating β as a configurable modeling parameter rather than a fixed physical constant. In SAGE, β governs the temporal curvature of calendar aging but does not alter the underlying temperature or state-of-charge stress factors.

S3.2 Cycle Aging Discretization.

Cycle aging is computed by integrating the instantaneous rate equation over each hourly timestep under the assumption of constant discharge power within an active dispatch block. For a fixed-power block, the incremental cycle-related capacity loss accumulated over a timestep Δt is proportional to the discharged energy during that interval, yielding a throughput-proportional damage contribution at reference conditions,

$$\Delta q_{\text{cyc}} = k_{\text{cyc}} \mid P_{\text{cap}}^{\text{batt}} \mid f_{T,\text{cyc}} \Delta t \tag{S10}$$

This discretization is consistent with grid-scale operation, where dispatch is commanded at constant power over multi-hour blocks, and avoids introducing current dependent microphysical detail that is not resolvable at the system level.

In SAGE we distinguish explicitly between grid-side power $P_{grid}(t)$ and battery-side power $P_{batt}(t)$. The grid-side quantity represents the power exchanged at the interconnection point and determines revenue and nameplate limits. The battery-side quantity represents the internal electrical power processed by the cells and is related to the grid-side value through the discharge efficiency, $P_{batt}(t) = P_{grid}(t)/\eta_{dis}(t)$. Cycle aging in Eq. (S10) is driven by battery-side power because electrochemical degradation scales with the internal current experienced by the electrodes. In contrast, the thermal model (Eq. 8 in the main text and Eq. S11 below) is written in grid-side form, $P_{heat} = P_{grid}(1/\eta_{dis} - 1)$, which is algebraically equivalent to the internal loss $P_{batt} - P_{grid}$. This separation ensures that electrochemical stress is computed from internal cell power while system-level heat generation and rated-power calibration remain consistent with externally defined efficiency and nameplate power.

S3.3 Formation Offset.

The calendar-aging power-law formulation yields a formally infinite instantaneous degradation rate at t = 0 for any β < 1. To ensure numerical stability, effective time is therefore initialized at a small positive value ($t_{\text{formation}}$ = 100 hours in the baseline configuration), representing pre-deployment aging during cell formation and logistics. This offset serves solely as a numerical regularization and has negligible influence on long horizon degradation trajectories, as it represents less than 0.05% of the total simulated lifetime in a 25-year horizon. Because the same offset is applied consistently across all simulations, it does not affect relative comparisons between scenarios or sensitivity results.

S3.4 Thermal Model Calibration and Constraints.

The thermal resistance coefficient K_T (°C/kW) is derived directly from the user-specified steady-state temperature rise at rated C/4 discharge, $\Delta T_{C/4}$, rather than fit to transient

data. This calibration establishes a lumped, system-level relationship between electrical power dissipation and cell temperature rise that is appropriate for long-horizon fleet simulations. The formulation intentionally abstracts detailed heat-transfer pathways and thermal inertia, focusing instead on steady-state operating limits and average thermal stress. Users modeling specific enclosures or cooling architectures may adjust $\Delta T_{C/4}$ to reflect alternative thermal designs without modifying the governing equations.

$$K_T = \frac{\Delta T_{C/4}}{P_{\text{rated}}\left(\frac{1}{\eta_{\text{ref}}} - 1\right)} \tag{S11}$$

The baseline configuration assumes $\Delta T_{C/4} = 2°\text{C}$ at rated C/4 discharge.

S3.4.1 Thermal Derating. To ensure $T_{cell} \leq T_{max}$, SAGE computes a thermally limited maximum power P_{therm} by inverting the steady-state thermal equation:

$$P_{\text{therm}} = \frac{T_{\text{max}} - T_{\text{amb}} - T_{\text{static}}}{K_T\left(\frac{1}{\eta_{\text{dis}}} - 1\right)} \tag{S12}$$

Physically, the numerator represents the thermal headroom (how many degrees the battery can warm up before hitting the limit), and the denominator represents the heating rate per kW of discharge. Dispatch power is strictly limited to min($P_{command}$, P_{therm}), ensuring safety constraints are respected during extreme weather.

S4. BASELINE MODEL CALIBRATION

The baseline degradation models introduced in the main text—the linear aging model and the throughput-only model—are included solely as reference cases to contextualize SAGE's physics-informed formulation. These models are not intended to represent physically complete descriptions of battery aging. Instead, they serve as simplified benchmarks against which differences in degradation trajectory shape, sensitivity to operating conditions, and failure timing can be evaluated. To enable meaningful

comparison, both baseline models are calibrated to match the end-of-life time of a reference SAGE simulation under identical dispatch, thermal, and outage conditions.

S4.1 Linear Model.

The linear baseline model assumes a constant degradation rate, yielding a strictly linear decline in state of health with elapsed time. This formulation intentionally neglects all dependence on temperature, state of charge, throughput, and operational history, and therefore does not represent a physically realistic aging mechanism. It is included solely as a minimal reference case to illustrate how deviations from linearity arise when aging is governed by stress-dependent physics rather than by elapsed time alone.

$$\mathrm{SOH}(t) = 1 - r \cdot t. \tag{S13}$$

rom your screenshot, S14 is:

$$r = \frac{1 - \mathrm{SoH}^{\mathrm{EOL}}}{t_{\mathrm{EOL,SAGE}}} \tag{$S14$}$$

S4.2 Throughput Model.

The throughput-only baseline model assumes degradation is proportional to cumulative discharged energy, independent of temperature, state of charge, or temporal operating patterns. The proportionality coefficient is calibrated such that the model reaches end of life at the same time as the reference SAGE simulation under baseline conditions. This calibration ensures that comparisons isolate differences in degradation trajectory shape rather than overall lifetime scaling. Agreement in end-of-life timing under this specific operating regime does not imply physical equivalence between the throughput model and SAGE. Instead, it reflects the near-linear accumulation of discharged energy with time under the baseline dispatch schedule, a property that does not hold under more heterogeneous or intermittent operating conditions.

$$\mathrm{SOH}(E) = 1 - k_E \cdot E_{\mathrm{cumulative}}. \tag{S15}$$

The calibration that guarantees $\mathrm{SoH}(E_{\mathrm{EOL,SAGE}}) = \mathrm{SoH}^{\mathrm{EOL}}$is:

$$k_E = \frac{1 - \mathrm{SoH}^{\mathrm{EOL}}}{E_{\mathrm{EOL,SAGE}}} \tag{S16}$$

(Units: $k_E$in MWh^{-1} if $E_{\mathrm{EOL,SAGE}}$is in MWh.)

where $E_{\mathrm{EOL,SAGE}}$ is the cumulative energy discharged by the reference asset at end-of-life. All baseline models were evaluated under identical dispatch, thermal, and outage conditions as SAGE, ensuring that differences in lifetime arise solely from degradation formulation rather than operational exposure.

S5. SENSITIVITY ANALYSIS DETAILS

This section summarizes the results of the one-at-a-time sensitivity analysis used to identify which physical uncertainties exert the strongest influence on fleet-level longevity metrics in the SAGE framework. Sensitivities are evaluated with respect to fleet-average lifespan, dispersion, and lower-tail outcomes, rather than individual asset trajectories, reflecting the population-level focus of the model. The results are intended to establish relative importance under the baseline operating and thermal management assumptions, rather than to provide site-independent or universally optimal design guidance.

S5.1 Parameter Sweep Configurations

Table S3 specifies the parameter sweep configurations used in the sensitivity analysis. For each configuration, a complete fleet simulation (100 assets over a 25-year horizon) was executed while holding all non-swept parameters fixed at their baseline values. Random seeds for environmental variability and fleet heterogeneity were held constant across sweeps to ensure that differences in outcomes reflect parameter perturbations rather than Monte Carlo noise.

Parameter	Baseline	Sweep Values	Units
sigma_manufacturing_quality	0.02	0.01; 0.02; 0.03; 0.05	—
vertical_temp_gradient_C	5	2; 5; 8	°C
E_a_cal	53000	45000; 53000; 60000	J/mol
E_a_cyc	35000	25000; 35000; 45000	J/mol
container_baseline	22	18; 22; 26; 30	°C
beta_cal	0.75	0.5; 0.6; 0.75	—
num_discharge_hours	4	2; 4; 6	h
SoH_EOL	0.7	0.6; 0.7; 0.8	—

S5.2 Detailed Simulation Results

Table S4 presents detailed fleet-level statistics for each parameter configuration, including mean lifespan, standard deviation, and lower-tail percentiles. These results form the basis for the regression and elasticity calculations described below.

Parameter	Value	Mean	Std	P10	P50	P90	Min	Max
E_a_cal	45,000	14.54	1.07	13.12	14.52	15.78	12.54	16.93
E_a_cal	53,000	14.55	1.12	13.07	14.52	15.85	12.46	17.03
E_a_cal	60,000	14.56	1.16	13.03	14.51	15.91	12.40	17.13
E_a_cyc	25,000	14.81	0.94	13.56	14.84	15.95	12.98	16.93
E_a_cyc	35,000	14.55	1.12	13.07	14.52	15.85	12.46	17.03
E_a_cyc	45,000	14.30	1.29	12.64	14.24	15.82	11.93	17.14
SoH_EOL	0.60	19.92	1.52	17.91	19.85	21.70	17.05	23.31
SoH_EOL	0.70	14.55	1.12	13.07	14.52	15.85	12.46	17.03
SoH_EOL	0.80	9.35	0.72	8.41	9.32	10.18	7.98	10.96
beta_cal	0.50	20.15	1.39	18.32	20.11	21.77	17.54	23.25
beta_cal	0.60	19.36	1.35	17.58	19.33	20.93	16.81	22.38
beta_cal	0.75	14.55	1.12	13.07	14.52	15.85	12.46	17.03
container_baseline	18.00	18.00	1.41	16.15	17.93	19.65	15.37	21.13
container_baseline	22.00	14.55	1.12	13.07	14.52	15.85	12.46	17.03
container_baseline	26.00	11.81	0.89	10.65	11.78	12.85	10.12	13.81
container_baseline	30.00	9.63	0.71	8.69	9.62	10.46	8.28	11.22
num_discharge_hours	2.00	18.29	1.45	16.39	18.22	19.97	15.59	21.53
num_discharge_hours	4.00	14.55	1.12	13.07	14.52	15.85	12.46	17.03
num_discharge_hours	6.00	14.94	1.15	13.44	14.89	16.28	12.79	17.53
sigma_manufacturing_quality	0.01	14.57	1.07	13.14	14.61	15.91	12.66	16.80
sigma_manufacturing_quality	0.02	14.55	1.12	13.07	14.52	15.85	12.46	17.03
sigma_manufacturing_quality	0.03	14.54	1.18	12.96	14.60	16.01	12.26	17.29
sigma_manufacturing_quality	0.05	14.51	1.37	12.72	14.54	16.20	11.87	17.79
vertical_temp_gradient_C	2.00	15.72	0.60	14.93	15.75	16.46	14.54	17.07
vertical_temp_gradient_C	5.00	14.55	1.12	13.07	14.52	15.85	12.46	17.03
vertical_temp_gradient_C	8.00	13.50	1.59	11.46	13.53	15.49	10.71	17.00

S5.3 Elasticity Definition

To enable comparison across parameters with different physical units and magnitudes, sensitivities are reported as local elasticities, defined as

$$\varepsilon = \frac{p_0}{L_0}\left(\frac{dL}{dp}\right)_{p_0}$$

where p_0 and L_0 denote the baseline parameter value and baseline mean fleet lifespan, respectively. Elasticities reported in this work are evaluated about the baseline configuration and summarize first-order sensitivity.

S5.4 Generalizability Considerations

The weak sensitivity to activation energies observed here is conditional on the baseline thermal regime. With average cell temperatures near 25 °C, the Arrhenius term ($1/T_{\text{ref}} - 1/T$) remains small, rendering lifespan relatively insensitive to uncertainty in activation energy. In hotter climates, where cells operate at sustained temperatures of 35–45 °C, the same parameter uncertainty would produce substantially larger lifespan variation.

Notably, $E_{a,\text{cyc}}$ exhibits approximately six times greater sensitivity than $E_{a,\text{cal}}$, despite calendar aging operating continuously. This reflects the higher cell temperatures experienced during active discharge relative to idle periods.

S6. COMPUTATIONAL PERFORMANCE

S6.1 Execution Time

Benchmarks were obtained on Google Colab (CPU runtime, October 2025 environment) with the following configuration: Intel® Xeon® CPU @ 2.20 GHz (8 logical CPUs; 4 cores × 2 threads), 54.8 GB RAM, Python 3.12.12, Linux x86_64 (kernel 6.6).

For the baseline configuration (1,000 assets simulated over 25 years; 219,000 hourly time steps per asset), a full fleet simulation completes in approximately 140 minutes of wall-clock time, corresponding to roughly 8 seconds per asset when executed sequentially.

Execution time scales approximately linearly with both fleet size and simulation duration. Because asset simulations are independent given the shared exogenous environment, runtime can be reduced proportionally through parallel execution across assets or across parameter cases. These results indicate that hourly-resolved, multi-decade fleet simulations at 1,000-asset scale are computationally tractable on commodity CPU hardware.

S6.2 Memory Usage

Memory requirements remain modest because the fleet simulation retains only summary statistics for each asset (e.g., lifespan and selected aggregates), while full hourly trajectories are exported only for designated diagnostic assets.

Fleet lifespan summaries require on the order of ~100 bytes per asset. Monthly diagnostic trajectories require approximately ~2 kB per simulated year per asset. Under typical study configurations (e.g., 100–1,000 assets over 25 years), total memory usage remains well within the capacity of standard desktop systems.

S6.3 Parallelization Opportunities

The current reference implementation executes asset simulations sequentially. However, because assets evolve independently conditional on shared environmental inputs, the fleet simulation is embarrassingly parallel. Substantial acceleration is therefore achievable through:

- **Multi-core CPU parallelization:** Python multiprocessing across assets or across parameter-sweep cases (expected several-fold speedup on modern multi-core systems).
- **Vectorized fleet updates:** Batch state updates across assets (not implemented in the current reference version).
- **Distributed parameter sweeps:** Independent sensitivity or scenario runs executed across multiple cloud instances.

These parallelization strategies do not alter model physics and can be implemented without modification to the degradation framework.

S7. CODE STRUCTURE

```
SAGE/

├── sage_params.csv                # Master parameter configuration (all physics + market inputs)

├── sensitivity_sweeps.csv         # Parameter sweep definitions for sensitivity analysis

│

├── SAGE.ipynb                     # Core simulation engine (physics + fleet simulation)

├── SAGE_visualizations.ipynb      # Figures for degradation validation (Figure 3)

├── SAGE_strawman.ipynb            # Baseline comparison models (Figure 5)

├── SAGE_sensitivity.ipynb         # Batch sensitivity runner (executes parameter sweeps)

├── SAGE_Sensitivity_Analysis.ipynb  # Tornado plot + SI tables (Figure 7, Tables S7–S9)

│

├── sensitivity_results_final/     # Output from parameter sweeps

│   ├── sage_fleet_lifespans_*.csv

│   ├── Fig_tornado_elasticity.png

│   ├── Fig_tornado_elasticity.pdf

│   └── sensitivity_run_log.csv

│

├── sage_single_asset_trajectory.csv # Single-asset monthly trajectory
```

All notebooks operate from the same parameter CSV and write structured CSV outputs, enabling full reproducibility of all figures and tables reported in the manuscript and Supporting Information.

Supplementary References

[S1] Keil, P., Jossen, A. Calendar aging of lithium-ion batteries. J. Electrochem. Soc. 163 (2016) A1872–A1880.

[S2] Schmalstieg, J., Käbitz, S., Ecker, M., Sauer, D.U. A holistic aging model for $Li(NiMnCo)O_2$ based 18650 lithium-ion batteries. J. Power Sources 257 (2014) 325–334.